

Fuzzy State-Driven Cross-Time Spatial Dependence Learning for Multivariate Time-Series Anomaly Detection

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Abstract—Cross-time spatial dependence (i.e., the interaction between different variables at different time points) is indispensable for detecting anomalies in multivariate time series, as certain anomalies may have time delays in their propagation from one variable to another. However, accurately capturing cross-time spatial dependence remains a challenge. Specifically, real-world time series usually exhibits complex and incomprehensible evolutions that may be compounded by multiple temporal states (i.e., temporal patterns, such as rising, fluctuating, and peak). These temporal states mix and overlap with each other and exhibit dynamic and heterogeneous evolution laws in different time series, making the cross-time spatial dependence extremely intricate and mutable. Therefore, a cross-time spatial graph network with fuzzy embedding is proposed to disentangle latent and mixing temporal states and exploit it to meticulously learn cross-time spatial dependence. First, considering that temporal states are diversiform and their mixing modes are unknown, we introduce a fuzzy state set to uniformly characterize potential temporal states and adaptively generate corresponding membership degrees to depict how these states mix. Further, we propose a cross-time spatial graph, quantifying similarities among fuzzy states and sensing their dynamic evolutions, to flexibly learn mutable cross-time spatial dependence. Finally, we design state diversity and temporal proximity constraints to ensure the differences among fuzzy states and the evolution continuity of fuzzy states. Experiments on real-world datasets show that the proposed model outperforms the state-of-the-art models.

Index Terms—Anomaly detection, cross-time spatial dependence, fuzzy state, multivariate time series, temporal state.

NOMENCLATURE

N	Number of variables.
T	Number of time points.
ϑ	Convolution kernels.
MF_q^n	Membership function.
$md_{t,q}^n$	Membership degree.
μ_q^n and σ_q^n	Center and width of membership function.
W, b	Weight parameter and bias.
$\mathcal{L}_V$	State diversity constraint.
$\mathcal{L}_C$	Temporal proximity constraint.
$\mathcal{L}_R$	Reconstruction constraint.
$\mathcal{L}_P$	Prediction constraint.

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C_1, C_2 , and C_3	Desired dimensions of output channel.
$\lambda_1, \lambda_2, \lambda_3$	Tradeoff coefficients.
ω_1, ω_2	Tradeoff coefficients.
ϕ_{score}	Anomaly score.
η	Threshold to estimate anomalies.
x_t	Values of N variables at time point t .
$x_{1:T}^n$	Values of T time points of variable n .
$\tilde{X}_{1:T}$	Input sample sequence.
$\hat{X}_{1:T}$	Reconstructed values for input sample sequence.
$\hat{x}_T$	Predicted values of time series at time point T .
$\hat{y}_t$	Anomaly label.
fmv_t^n	Membership vector.
$\text{FEX}_{1:T}$	Fuzzy embedding data for input sample sequence.
$\text{CTSG}_{t:t+1}$	Cross-time spatial graph.
$A_{t:t+1}$	Normalized adjacency matrix.

I. INTRODUCTION

MULTIVARIATE time-series anomaly detection has been extensively developed and researched across diverse domains, aiming at finding abnormal conditions to ensure system security and reduce financial loss [1]. Since anomaly data are rare and difficult to gather, a popular and promising paradigm in multivariate time-series anomaly detection is to leverage unsupervised methods to learn specific patterns of time series to recognize underlying anomalies [2]. Nowadays, various unsupervised methods have been proposed and realized fruitful progress, such as Gaussian mixture model [3] and isolation forest (IF) [4].

However, the above methods remain limited ability because they do not consider crucial spatiotemporal dependencies in multivariate time series [5], [6]. Unlike other data (e.g., image and text), spatiotemporal dependencies are the distinguishing characteristic of multivariate time series [7]. As shown in Fig. 1(a), spatiotemporal dependencies are extremely complicated because they contain three parts, dubbed temporal dependence (cross-time interactions within the same variable), spatial dependence (interactions between different variables at the same time point), and cross-time spatial dependence (interactions between different variables at different time points). Vast studies have shown that accurately capturing spatiotemporal dependencies is vital to learning specific patterns of time series for detecting anomalies [8], [9].

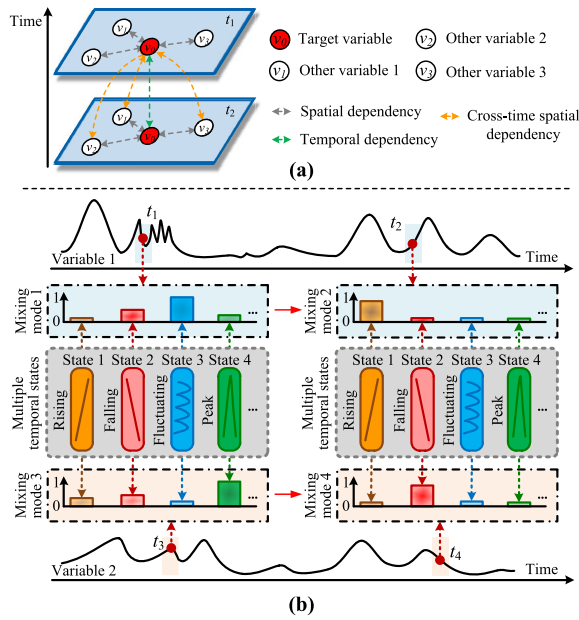


Fig. 1. Example of complex spatiotemporal dependencies and latent temporal states of time series. (a) Three different dependencies in spatiotemporal dependencies. (b) Multiple temporal states mix and overlap with each other, and present dynamic and heterogeneous evolution laws.

To this end, various deep learning-based studies have been proposed to mine spatiotemporal dependencies and gained remarkable momentum [10], [11]. For instance, transformer model [1], [12] and long short-term memory (LSTM) network [13] focus on capturing temporal dependence across the time dimension, while graph neural network (GNN) [14], [15] aims at explicitly modeling spatial dependence across the variable dimension. Unfortunately, these studies may suffer from false negative problems as the captured temporal dependence and spatial dependence are ineffective for detecting certain latent time-delay anomalies. In real-world industrial processes, the time-delay phenomenon in the propagation of anomalies is common because the effect of one entity (e.g., device or machine) may take a certain time to spread to another. One decent scheme for detecting time-delay anomalies is to exploit the cross-time spatial dependence because it can effectively reflect the intervariable influences at different time points, thus helping to model the propagation mechanism of time-delay anomalies.

However, modeling cross-time spatial dependence is still a challenging task as its intricateness and mutability. These characteristics may be induced by multiple temporal states hidden in time-series evolutions. Specifically, as shown in Fig. 1(b), time series of variable 1 and variable 2 present complex and incomprehensible evolutions, where multiple temporal states (or referred to temporal patterns) mix and overlap with each other, e.g., state 1 (rising), state 2 (falling), state 3 (fluctuating), and state 4 (peak) [16], [17]. It can be seen that latent temporal states show the specific mixing mode in each evolution phase of time series, in which different temporal states have different influence degrees on time-series evolutions. For example, assuming that variable 1 represents the altitude of an airplane, state 1 (rising) may exert larger influences during the take-off phase (roughly located at t_2).

In comparison, state 3 (fluctuating) influences more when encountering unstable airflow (roughly located at t_1). Moreover, it can also be found that temporal states exhibit dynamic and heterogeneous evolution laws in different time series (e.g., from mixing mode 1 to mixing mode 2 in variable 1 and from mixing mode 3 to mixing mode 4 in variable 2), thus making cross-time spatial dependence complex and mutable and further increasing its modeling difficulty.

In complex real scenarios, temporal states are usually diversiform and their mixing modes are often unknown (the influences from temporal states on time-series evolutions are difficult to quantify by precise and hard binary values). In light of this, we extend the analysis of time series from a 1-D space to a high-dimensional fuzzy space, aiming to construct a fuzzy state set (containing multiple fuzzy states) to uniformly characterize diversiform temporal states and generate soft membership degrees to portray their mixing modes. Through this manner, we can break the bottleneck of limited representation ability of 1-D time series and disentangle mixing temporal states, which helps to sense the mutability of cross-time spatial dependence and facilitate its fine-grained modeling, thus boosting the ability to discover time-delay anomalies.

Technically, we propose a novel cross-time spatial graph network with fuzzy embedding (FE-CTSNet), which consists of a fuzzy embedding method called time series to membership vector (TS2MVec), and a cross-time spatial graph (CTSG). TS2MVec constructs multiple fuzzy states and each of them represents a type of temporal state. Then, each time-series data point is transformed into a membership vector containing a series of membership degrees, where each membership degree indicates the degree to which the corresponding fuzzy state influences time-series evolutions. Based on the generated membership degrees, CTSG can quantify the similarity between fuzzy states of any two time-series data points, thereby flexibly learning cross-time spatial dependence. Moreover, membership degrees can adaptively change along with dynamic evolutions of time series, thereby enabling CTSG to sense the mutability of cross-time spatial dependence. Finally, to strengthen the ability of fuzzy states to characterize potential temporal states, we design state diversity and temporal proximity constraints, which not only make different fuzzy states have differences, but also ensure the evolution continuity of fuzzy states.

The main contributions of this study are threefold.

- 1) We propose a TS2MVec to construct a fine-grained fuzzy state set to uniformly characterize latent temporal states and adaptively generate corresponding membership degrees to depict their mixing modes. To the best of authors' knowledge, it is the first time to realize the disentanglement for mixing temporal states from a fuzzy embedding perspective.
- 2) We propose a CTSG to meticulously learn the cross-time spatial dependence. Different from the existing graph structures, it not only realizes flexible representations for cross-time spatial dependence, but also senses its mutability by capturing dynamic evolutions of fuzzy states.

TABLE I
DIFFERENCES BETWEEN FE-CTSNET AND OTHER METHODS

Models		Spatiotemporal dependencies		
		SD	TD	CTSD
Machine learning	IF [4]	×	×	×
	PCA [18]	✓	×	×
	SFA [20]	✓	✓	×
	CVA [21]	✓	✓	×
Deep learning	TranAD [12]	×	✓	×
	Anomaly Transformer [1]	×	✓	×
	OmniAnomaly [24]	×	✓	×
	AAU-Net [25]	×	✓	×
	DAGMM [27]	✓	×	×
	SG-AE [28]	✓	×	×
	GDN [26]	✓	×	×
	USAD [29]	✓	✓	×
	TopoMAD [32]	✓	✓	×
	MSCRED [30]	✓	✓	×
	MTAD-GAT [31]	✓	✓	×
FE-CTSNet		✓	✓	✓

*SD: spatial dependency, TD: temporal dependency, CTSD: cross-time spatial dependency.

- 3) We design state diversity and temporal proximity constraints, which can guarantee the differences among fuzzy states and the evolution continuity of fuzzy states, thereby further improving its ability to characterize temporal states.

II. RELATED WORK

A. Multivariate Time-Series Anomaly Detection

Unsupervised paradigms have been widely studied in the time-series anomaly detection field. Some popular methods focus on capturing complex spatiotemporal dependencies to learn specific patterns of time series and then detect latent anomalies. As shown in Table I, machine learning methods contain principal component analysis (PCA) [18], slow feature analysis (SFA) [19], [20], and canonical variate analysis (CVA) [21]. They aim to obtain low-dimensional latent variables from multivariate time series to represent its important information and then design customized indicators for latent variable spaces to detect anomalies. Among them, PCA considers spatial dependence while SFA and CVA can exploit both spatial dependence and temporal dependence. Although they have gained meaningful anomaly detection results, their extracted features are linear and thus not sufficient to perform well in complicated nonlinear scenarios.

At present, in unsupervised anomaly detection studies, deep learning methods have emerged frequently due to their superior ability to generate multilevel and nonlinear feature representations [22], [23]. On the one hand, some models focused on mining temporal dependence: such as LSTM-based models [24], autoencoder-based models [25], and transformer-based models [1], [12]. The first two types of models were only good at modeling short-range dependencies, while transformer-based models can use self-attention to capture reliable long-range dependencies, but with higher computational complexity. On the other hand, some models mainly modeled

spatial dependence and used it to discover anomalies, such as GNN-based models [26] and autoencoder-based models [27], [28]. Further, some other studies [29], [30], [31], [32] simultaneously considered the above two dependencies to promote anomaly detection results.

However, it can be seen from Table I that the existing unsupervised studies neglect the cross-time spatial dependence, making it hard to discover certain latent time-delay anomalies. In comparison, FE-CTSNet can model complete spatiotemporal dependencies (especially, cross-time spatial dependence), thereby realizing satisfactory anomaly detection results.

B. Membership Functions in Fuzzy Logic

Fuzzy logic and probability theory are two different reasoning methods [33]. Specifically, fuzzy logic can generate the membership degree to represent the degree to which an object has an abstract attribute (i.e., state) (generally without randomness) [34], [35]. Probability theory can generate the probability value to represent the likelihood of an object happening (generally with randomness) [36]. For each latent temporal state, membership degree portrays the degree of its influence on time-series evolutions, while probability value portrays the likelihood that it can influence time-series evolutions. Intuitively, fuzzy logic is more appropriate for this study since the influence of temporal states on time series has actually happened and directly induced its evolutions.

Recently, a promising method is to combine membership functions in the fuzzy logic with neural networks. For example, Deng et al. [37] used the Gaussian function as the membership function to generate fuzzy representations, which reduced the uncertainties in original data and showed its superiority in classification tasks. Ebadzadeh and Salimi-Badr [38] introduced a shapeable membership function with an adjustable shape and proposed interpretable correlated-contours fuzzy rules to realize better function approximation. All these studies use membership functions to generate fuzzy representations and automatically learn optimal parameters of membership functions, which is conducive to enhancing the representation of complex data.

III. METHODOLOGY

A. Problem Statement

In this study, we focus on introducing a fuzzy state set to characterize diversiform and mixing temporal states and exploit it to learn mutable cross-time spatial dependence for identifying potential time-delay anomalies. To formalize this problem, some key concepts are defined as follows.

Definition 1 (Task Description): A multivariate time series can be denoted as $X = \{x_1, x_2, \dots, x_\tau\} \in \mathbb{R}^{N \times \tau}$, where N is the number of variables and τ is the sequence length of X . $x_t = \{x_t^1, x_t^2, \dots, x_t^N\} \in \mathbb{R}^N$ is an N -dimensional sample (i.e., N variable values at time point t). Our goal is to predict the anomaly label of x_t , i.e., 1 (abnormal) or 0 (normal).

Definition 2 (Temporal States and Fuzzy States): We refer to mixing and overlapping temporal patterns involved in time-series evolutions as temporal states. Since temporal states are diversiform and their mixing modes are unknown,

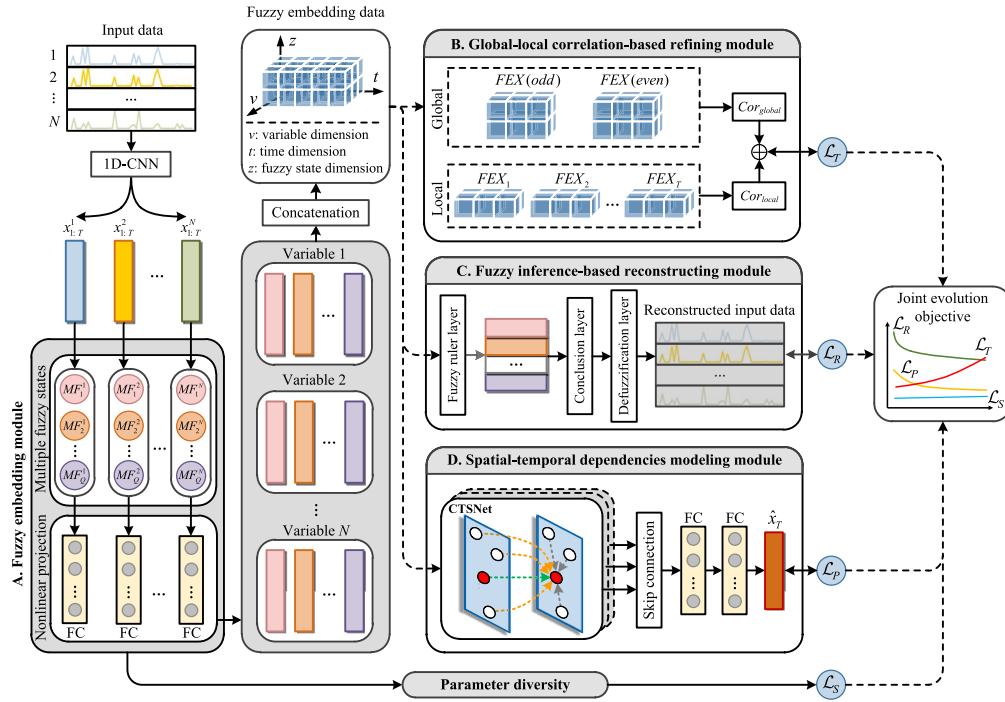


Fig. 2. Overall framework of FE-CTSNet. Module A is dedicated to generating fuzzy embedding data and ensuring its diversity, modules C and D are dedicated to using fuzzy embedding data from module A to achieve anomaly detection, and module B, as an auxiliary module, is dedicated to endowing the fuzzy embedding data from module A with temporal dependence and further facilitating the performance of modules C and D. These modules work together to gain the FE-CTSNet with superior performance.

we depict them by a fuzzy state set, in which each fuzzy state characterizes a type of temporal state. Then, we map each time-series data point into a membership vector (consisting of multiple membership degrees), where each membership degree indicates the degree to which the corresponding fuzzy state influences time-series evolutions.

The main notations in this study are summarized in Nomenclature.

B. Overall Framework

As shown in Fig. 2, FE-CTSNet contains four modules and exploits a joint evolution objective to co-optimize all the modules and guide end-to-end training. The process and contribution of each module are illustrated as follows.

1) *Fuzzy Embedding Module (Module A)*: We propose a new TS2MVec to build a fuzzy state set and design a new state diversity constraint to ensure differences among different fuzzy states, thereby enabling fuzzy states to characterize potential temporal states from as many perspectives as possible. Then, we introduce multiple membership functions to transform each time-series data point into a membership vector (containing a series of membership degrees). Finally, we combine all membership vectors along the time axis to generate fuzzy embedding data to effectively describe the evolution law of time series and is used as input for modules B–D.

2) *Global-Local Correlation-Based Refining Module (Module B)*: Considering that temporal states natively have temporal dependence, we design a new temporal proximity constraint to endow all membership vectors in fuzzy embedding data with temporal dependence, thus ensuring the evolution continuity of fuzzy states and improving its characterization ability for temporal states.

3) *Fuzzy Inference-Based Reconstructing Module (Module C)*: We design a customized fuzzy inference-based defuzzification method (containing fuzzy rule, conclusion, and defuzzification layers) for fuzzy embedding data to generate reconstructed time series as close as possible to the original time series.

4) *Spatiotemporal Dependencies Modeling Module (Module D)*: We propose a new CTSG to exploit fuzzy embedding data to learn complete spatiotemporal dependencies (i.e., spatial dependence, temporal dependence, and cross-time spatial dependence). Moreover, we design a cross-time spatial graph-based network (CTSNet) adapted to CTSG, exploiting deep hierarchical architectures to mine long-term spatiotemporal dependencies to obtain accurate forecasting results.

C. Fuzzy Embedding Module

In this module, we map all time series from a low-dimensional space (1-D) into a common high-dimensional fuzzy space, where each dimension corresponds to a fuzzy state. Specifically, we build a fuzzy state set (containing some membership functions) to transform each time-series data point into a series of membership degrees, and each of them indicates the degree to which the corresponding fuzzy state influences time-series evolutions. Then, all membership degrees of each time-series data point are concatenated to generate a membership vector representing its mixing mode of fuzzy states. In addition, considering that different variables usually exhibit heterogeneous evolution laws, we assign fuzzy state sets with different initial parameter values for different variables, which can make TS2MVec to capture the particular evolution law of each variable.

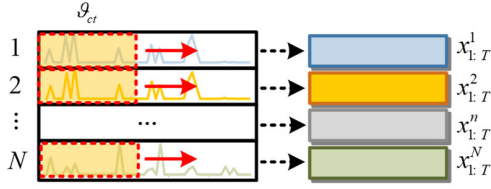


Fig. 3. Schematic of integrating the contextual information. Each yellow dotted box denotes a convolution kernel.

This module contains two stages: integration of contextual information and TS2MVec.

1) *Integration of Contextual Information*: Assume that $X_{1:T} \in \mathbb{R}^{N \times T}$ denotes the input sample sequence, where N and T are the number of variables and time points, respectively. As shown in Fig. 3, to make the fuzzy embedding method not be an isolated single-point embedding method, the contextual information along the time dimension is first integrated into each variable value

$$X'_{1:T} = X_{1:T} * \vartheta_{ct} \quad (1)$$

where $X'_{1:T} \in \mathbb{R}^{N \times T}$ and $\vartheta_{ct} \in \mathbb{R}^{1 \times \delta \times 1}$ denote the output data and the contextual convolution kernel, respectively. δ denotes the time span of contextual information.

In particular, the padding operation is performed during the convolution process to ensure the size consistency between $X_{1:T}$ and $X'_{1:T}$. Then, $X'_{1:T}$ is segmented into N feature slices $x'_{1:T}, x'_{1:T}, \dots, x'_{1:T} \in \mathbb{R}^T$ along the variable dimension, and then, TS2MVec is carried out on each feature slice $x'_{1:T}$ ($1 \leq n \leq N$), respectively.

2) *TS2MVec*: Following the classic research on fuzzy neural network (FNN) [37], [39], we adopt the Gaussian function as the membership function because it has the advantages of differentiability and continuity. We construct different fuzzy state sets for different variables and each fuzzy state set contains Q different membership functions (denoting Q fuzzy states). As shown in Fig. 4, taking the n th ($1 \leq n \leq N$) variable as an example, its variable values at T time points $x^n_{1:T}$ are all input into the n th fuzzy state set that contains Q membership functions, as shown in the following equation:

$$\begin{aligned} \text{md}_{t,q}^n &= \text{MF}_q^n(x_t^n) \\ &= e^{-\frac{(x_t^n - \mu_q^n)^2}{(\sigma_q^n)^2}}, \quad 1 \leq t \leq T, \quad 1 \leq q \leq Q \end{aligned} \quad (2)$$

where $x_t^n \in \mathbb{R}$ denotes the time-series data point at t th time point. $\text{md}_{t,q}^n$ and MF_q^n denote the membership degree (i.e., the influence degree of q th fuzzy state) and membership function, respectively. μ_q^n and σ_q^n denote the center and the width of MF_q^n , respectively, which can be adaptively updated during model training.

Then, the outputs of Q membership functions (i.e., Q membership degrees) are concatenated together to constitute the membership vector for $x_t^n \in \mathbb{R}$. Moreover, we use the nonlinear mapping to obtain the membership vector with stronger characterization ability, as shown in the following equations:

$$\text{md}_t^n = \text{md}_{t,1}^n || \text{md}_{t,2}^n || \dots || \text{md}_{t,Q}^n \quad (3)$$

$$\text{fmv}_t^n = \tanh(\text{md}_t^n W_t^n + b_t^n) \quad (4)$$

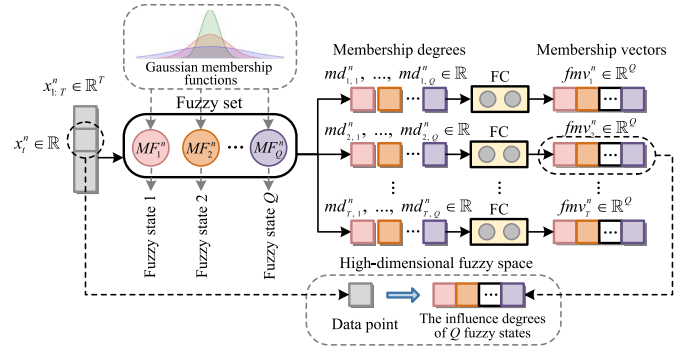


Fig. 4. Schematic of TS2MVec for n th variable. Q membership functions can output Q membership degrees for each time-series data point to form a membership vector to characterize Q fuzzy states.

where $\text{md}_t^n \in \mathbb{R}^Q$ denotes the joint of membership degrees. $\text{fmv}_t^n \in \mathbb{R}^Q$ denotes the final membership vector that can characterize the mixing mode of Q fuzzy states for $x_t^n \in \mathbb{R}$. $W_t^n \in \mathbb{R}^{Q \times Q}$ and $b_t^n \in \mathbb{R}^Q$ denote the weight parameter and bias, respectively. $||$ denotes the concatenation operation.

Subsequently, the membership vectors corresponding to T variable values in $x^n_{1:T}$ are concatenated together to gain the fuzzy embedding data $\text{fmv}_{1:T}^n \in \mathbb{R}^{T \times Q}$ for $x^n_{1:T}$, as shown in the following equation:

$$\text{fmv}_{1:T}^n = \text{fmv}_1^n || \text{fmv}_2^n || \dots || \text{fmv}_T^n. \quad (5)$$

Finally, the fuzzy embedding data of N feature slices $\{x_{1:T}^1, x_{1:T}^2, \dots, x_{1:T}^N\}$ are concatenated together to obtain the final fuzzy embedding data $\text{FEX}_{1:T} \in \mathbb{R}^{N \times T \times Q}$ for the input sample sequence $X_{1:T} \in \mathbb{R}^{N \times T}$, as shown in the following equation:

$$\text{FEX}_{1:T} = \text{fmv}_{1:T}^1 || \text{fmv}_{1:T}^2 || \dots || \text{fmv}_{1:T}^N. \quad (6)$$

To make different fuzzy states that can characterize temporal states with different semantics, we design a state diversity constraint to guarantee parameter diversity of Gaussian functions to make fuzzy states have differences. It can calculate the variance values of the centers of all Gaussian functions in N fuzzy state sets and maximize their sum, as shown in the following equation:

$$\mathcal{L}_S = -\frac{1}{N} \sum_{n=1}^N \text{Var}(\mu_1^n, \mu_2^n, \dots, \mu_Q^n) \quad (7)$$

where $\mathcal{L}_S$ and Var denote the state diversity constraint and variance function, respectively.

D. Global-Local Correlation-Based Refining Module

Due to temporal states natively having temporal dependence, we design a temporal proximity constraint to make fuzzy embedding data possess temporal dependence and evolve continuously.

1) *Global Perspective*: As shown in the top of Fig. 2 (module A), the interleaved downsampling [40] is performed on fuzzy embedding data, which separates odd and even elements along the time dimension to obtain two subsequences $\text{FEX}(\text{odd})$ and $\text{FEX}(\text{even})$, as shown in the following equation:

$$\text{DS}(\text{FEX}_{1:T}) = [\text{FEX}(\text{odd}), \text{FEX}(\text{even})] \quad (8)$$

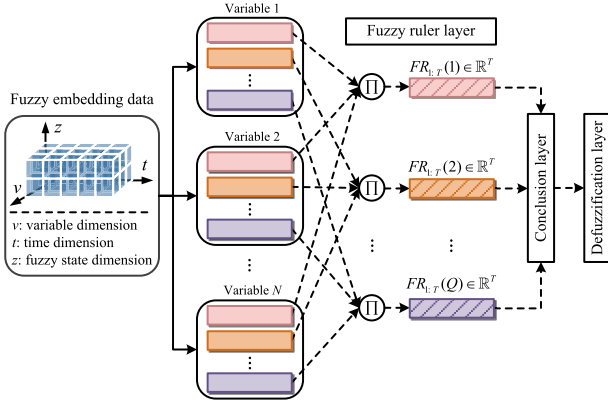


Fig. 5. Fuzzy inference uses a fuzzy ruler layer, conclusion layer, and defuzzification layer to reconstruct time series.

where DS denotes the interleaved downsampling operation.

Then, FEX(odd) and FEX(even) are flattened to calculate the correlation between them, as shown in the following equation:

$$\text{Cor}_{\text{global}} = \text{Cossim}(\text{Ft}[\text{FEX}(\text{odd})], \text{Ft}[\text{FEX}(\text{even})]) \quad (9)$$

where $\text{Cor}_{\text{global}}$ denotes the global correlation. $\text{Ft}[\cdot]$ denotes the flattening operation. Cossim denotes the cosine similarity.

Due to the trend and closeness properties of time-series data, two subsequences obtained by interleaved downsampling still possess temporal dependence of original time-series data [40], [41]. Therefore, if the global correlation between two subsequences [i.e., FEX(odd) and FEX(even)] is higher, the stronger temporal dependence characteristic of fuzzy embedding data will be resulted.

2) *Local Perspective*: As shown in the bottom of Fig. 2 (module A), the cosine similarity method is used to successively calculate the correlation between fuzzy embedding data belonging to adjacent time points, as shown in the following equation:

$$\text{Cor}_{\text{local}} = \frac{1}{T-1} \sum_{t=1}^{T-1} \text{Cossim}(\text{Ft}[\text{FEX}_t], \text{Ft}[\text{FEX}_{t+1}]) \quad (10)$$

where $\text{Cor}_{\text{local}}$ denotes the local correlation. FEX_t denotes the fuzzy embedding data at time point t .

Finally, the global correlation and local correlation are synthetically considered to build the temporal proximity constraint $\mathcal{L}_T$, which is defined by the following equation:

$$\mathcal{L}_T = \frac{(\text{Cor}_{\text{global}} + \text{Cor}_{\text{local}})}{2}. \quad (11)$$

E. Fuzzy Inference-Based Reconstructing Module

In this section, the fuzzy inference method is introduced to generate the reconstructed time series as close as possible to the original time series. As shown in Fig. 5, the fuzzy inference method contains a fuzzy rule layer, a conclusion layer, and a defuzzification layer.

The description of them is elaborated as follows.

1) *Fuzzy Rule Layer*: The number of neurons in this layer is Q , which is equal to the number of membership functions in a fuzzy state set. Each neuron is regarded as a fuzzy rule,

and its output indicates the firing strength of the fuzzy rule in response to the original time-series data [35], [42]. We use the product T -norm operator on the fuzzy embedding data, as shown in the following equation:

$$\text{FR}_{1:T}(q) = \prod_n^N \text{FEX}_{1:T}^n(q), \quad 1 \leq q \leq Q \quad (12)$$

where $\text{FR}_{1:T}(q) \in \mathbb{R}^T$ denotes the firing strength of q th neuron (q th fuzzy rule). $\text{FEX}_{1:T}^n(q) \in \mathbb{R}^T$ denotes the membership vector corresponding to n th variable and q th membership function.

2) *Conclusion Layer*: This layer normalizes the firing strengths of all fuzzy rules and concatenates them together, as shown in the following equations:

$$\text{FRN}(q) = \frac{\text{FR}_{1:T}(q)}{\sum_{i=1}^Q \text{FR}_{1:T}(i)} \quad (13)$$

$$\text{FRN}_{\text{all}} = \text{FRN}(1) || \text{FRN}(2) || \dots || \text{FRN}(Q) \quad (14)$$

where $\text{FRN}(q) \in \mathbb{R}^T$ denotes the normalized firing strength of q th neuron (q th fuzzy rule). $\text{FRN}_{\text{all}} \in \mathbb{R}^{T \times Q}$ denotes the final firing strength matrix.

3) *Defuzzification Layer*: This layer uses the nonlinear mapping to perform defuzzification operation for generating the reconstructed time-series data, as shown in the following equation:

$$\tilde{X}_{1:T} = \tanh(\text{FRN}_{\text{all}} W_R + b_R) \quad (15)$$

where $W_R \in \mathbb{R}^{Q \times N}$ and $b_R \in \mathbb{R}^N$ denote the weight parameter and bias, respectively. $\tilde{X}_{1:T} \in \mathbb{R}^{T \times N}$ denotes the reconstructed time series. Moreover, its shape is converted to $\mathbb{R}^{N \times T}$ to ensure consistency with the shape of the original time-series data.

Finally, the loss function $\mathcal{L}_R$ is built, which is defined by the following equation:

$$\mathcal{L}_R = \frac{1}{N \times T} \|X_{1:T} - \tilde{X}_{1:T}\|_2^2 \quad (16)$$

where N and T denote the number of variables and time points, respectively. $X_{1:T}$ and $\tilde{X}_{1:T}$ denote the original values and reconstructed values, respectively. $\|\cdot\|_2$ denotes the L2-norm.

F. Spatiotemporal Dependencies Modeling Module

This module adopts a prediction-oriented framework, using the data at historical time points $(1, 2, \dots, T-1)$ to predict the data at future time point T , and thus, the input of this module is a partial of fuzzy embedding data, i.e., $\text{FEX}_{1:T-1} \in \mathbb{R}^{N \times (T-1) \times Q}$.

1) *CTSG Generation*: As shown in Fig. 6, CTSG uses the membership vectors generated by TS2MVec to calculate similarities among all time-series data points (including all variables) across two adjacent time points. Then, all the similarities are aggregated to explicitly represent complete spatiotemporal dependencies (i.e., spatial dependence, temporal dependence, and cross-time spatial dependence).

The generation process of CTSG is described as follows.

First, $\text{FEX}_{1:T-1}$ is split into data slices $\text{FEX}_{1:2}, \text{FEX}_{2:3}, \dots, \text{FEX}_{T-2:T-1} \in \mathbb{R}^{N \times 2 \times Q}$, and then,

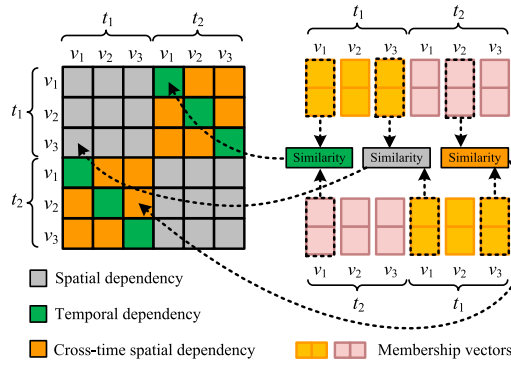


Fig. 6. Example of CTSG generation.

the corresponding CTSG is generated for each data slice. For $FEX_{t:t+1} \in \mathbb{R}^{N \times 2 \times Q}$ ($1 \leq t \leq T-2$), it is first reshaped as $FEX_{t:t+1} \in \mathbb{R}^{2N \times Q}$, which contains the membership vectors of $2N$ variables (N variables at time point t and N variables at time point $t+1$). Then, the generation process can be formulated by the following equation:

$$CTSG_{t:t+1}(i, j) = \text{Cossim}(FEX_{t:t+1}(i), FEX_{t:t+1}(j))$$

$$i, j = 1, 2, \dots, 2N \quad (17)$$

where i and j denote i th and j th nodes in CTSG, respectively. $FEX_{t:t+1}(i)$ and $FEX_{t:t+1}(j)$ denote the membership vectors of the i th and j th variables, respectively. Moreover, sparse operation is performed to preserve the most important correlations to obtain the final $CTSG_{t:t+1} \in \mathbb{R}^{2N \times 2N}$.

In particular, all the membership vectors change adaptively as the time series evolves, which endows the CTSG with the ability to learn mutable spatiotemporal dependencies (especially, cross-time spatial dependence). In CTSG, we only consider the spatiotemporal dependencies within two adjacent time points because build larger size (e.g., $3N \times 3N$ or $4N \times 4N$) to characterize longer time spans will bring higher computational costs. Thus, we unite multiple CTSGs to construct a deep hierarchical architecture (i.e., CTSNet), which uses the stacking of graph convolution layers to capture long-term spatiotemporal dependencies.

2) *CTSNet*: As shown in Fig. 7, CTSNet possesses a deep hierarchical architecture, which contains multiple stacked layers to capture long-term spatiotemporal dependencies, and employs the skip connection to obtain high-level feature representations. In each stacked layer, multiple independent mix-hop graph neural networks (MHGNNs) are customized for multiple data slices $FEX_{1:2}, FEX_{2:3}, \dots, FEX_{T-2:T-1}$, and these MHGNNs can carry out in parallel to enhance the model efficiency. Specifically, each MHGNN exploits multiple graph convolutional structures to extract features and introduces the activation function gated linear unit (GLU) to conduct nonlinear transformations and then aggregate the features of all graph convolutional structures by a max-pooling layer.

Technically, the reshaped data slice $FEX_{t:t+1} \in \mathbb{R}^{2N \times Q}$ and its corresponding $CTSG_{t:t+1} \in \mathbb{R}^{2N \times 2N}$ are input into a mix-hop graph convolutional network (GCN), as shown in the following equations:

$$A_{t:t+1} = D_{t:t+1}^{-\frac{1}{2}} CTSG_{t:t+1} D_{t:t+1}^{-\frac{1}{2}} \quad (18)$$

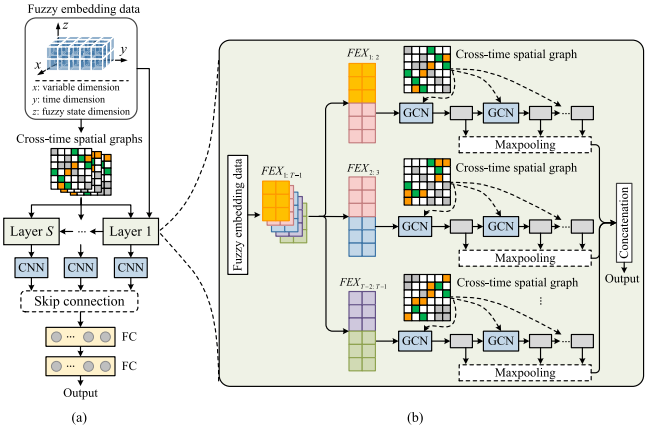


Fig. 7. Hierarchical architecture of the proposed CTSNet. (a) CTSNet adopts S layers to extract hierarchical feature representations and uses skip connection to fuse them. (b) In each layer in CTSNet, multiple mix-hop GCNs are conducted parallelly and independently.

$$H_{t:t+1}^l = \begin{cases} (A_{t:t+1} FEX_{t:t+1} W_1^l + b_1^l) \\ \odot \sigma(A_{t:t+1} FEX_{t:t+1} W_2^l + b_2^l), & l = 1 \\ (A_{t:t+1} H_{t:t+1}^{l-1} W_1^l + b_1^l) \odot \sigma(A_{t:t+1} H_{t:t+1}^{l-1} W_2^l + b_2^l), & l = 2, 3, \dots, L \end{cases} \quad (19)$$

where $D_{t:t+1} \in \mathbb{R}^{2N \times 2N}$ is the degree matrix of $CTSG_{t:t+1}$. $A_{t:t+1} \in \mathbb{R}^{2N \times 2N}$ is the normalized adjacency matrix of $CTSG_{t:t+1}$. $W_1^l, W_2^l \in \mathbb{R}^{Q \times C_1}$, and $b_1^l, b_2^l \in \mathbb{R}^{C_1}$ are the parameters in the activation function GLU [43]. C_1 is the desired output channel dimension. L denotes the number of layers in mix-hop GCN. $H_{t:t+1}^l \in \mathbb{R}^{2N \times C_1}$ denotes the output of the l th layer, which contains two parts of spatiotemporal dependencies features (i.e., $H_t^l \in \mathbb{R}^{N \times C_1}$ at time point t and $H_{t+1}^l \in \mathbb{R}^{N \times C_1}$ at time point $t+1$).

Next, we aggregate them to obtain a consistent representation of spatiotemporal dependencies over this time period (t and $t+1$)

$$HA_{t:t+1}^l = \text{Agg}(H_t^l, H_{t+1}^l) \quad (20)$$

where $HA_{t:t+1}^l \in \mathbb{R}^{N \times C_1}$ denotes the output. $\text{Agg}(\cdot)$ denote the elementwise addition.

To eliminate useless information and retain important information in L layers, the max-pooling is performed, which is formulated by the following equation:

$$HM_{t:t+1} = \text{Maxpooling}(HA_{t:t+1}^1, HA_{t:t+1}^2, \dots, HA_{t:t+1}^L). \quad (21)$$

Then, the outputs of all mix-hop GCNs are concatenated to form the final output $U_{\text{CTSNet}}^{(1)} \in \mathbb{R}^{N \times (T-2) \times C_1}$, as shown in the following equation:

$$U_{\text{CTSNet}}^{(1)} = HM_{1:2} || HM_{2:3} || \dots || HM_{T-2:T-1}. \quad (22)$$

The outputs of S stacked layers are fused by the skip connection to capture long-term spatiotemporal dependencies, as shown in the following equation:

$$U_{\text{all}} = \sum_{s=1}^S U_{\text{CTSNet}}^{(s)} * \vartheta_s \quad (23)$$

where $U_{\text{CTSN}}^{(s)} \in \mathbb{R}^{N \times (T-1-s) \times C_1}$ denotes the output of the s th layer in CTSNet. $\vartheta_s \in \mathbb{R}^{1 \times (T-1-s) \times C_2}$ denotes the s th convolution kernel. C_2 is the desired output channel dimension. $U_{\text{all}} \in \mathbb{R}^{N \times C_2}$ denotes the high-level feature representations.

Subsequently, two fully connected layers are used to refine the feature information in U_{all} and output the predicted results $\hat{x}_T \in \mathbb{R}^N$, as shown in the following equation:

$$\hat{x}_T = g(U_{\text{all}} W_3 + b_3) W_4 + b_4 \quad (24)$$

where $W_3 \in \mathbb{R}^{C_2 \times C_3}$, $W_4 \in \mathbb{R}^{C_3 \times 1}$ denote the weight parameters. $b_3 \in \mathbb{R}^{C_3}$, $b_4 \in \mathbb{R}$ denote the biases (C_3 denotes the number of neurons in the first fully connected layer). g denotes the activation function rectified linear unit (ReLU) [44].

Finally, the loss function $\mathcal{L}_P$ is built, which is defined by the following equation:

$$\mathcal{L}_P = \frac{1}{N} \|x_T - \hat{x}_T\|_2^2 \quad (25)$$

where N denotes the number of variables. x_T and $\hat{x}_T$ denote the real and predicted values at time point T , respectively.

G. Joint Evolution Objective

In this section, a joint evolution objective is designed to train the FE-CTSNet, including state diversity constraint $\mathcal{L}_S$, temporal proximity constraint $\mathcal{L}_T$, reconstruction constraint $\mathcal{L}_R$, and prediction constraint $\mathcal{L}_P$, as shown in the following equation:

$$\mathcal{L}_J = \lambda_1 \mathcal{L}_S + \frac{\lambda_2^{\text{epo}}}{2} \mathcal{L}_T + \frac{\lambda_3^{\text{epo}}}{2} \mathcal{L}_R + \left(1 - \lambda_1 - \frac{\lambda_2^{\text{epo}}}{2} - \frac{\lambda_3^{\text{epo}}}{2}\right) \mathcal{L}_P \quad (26)$$

where λ_1 , λ_2 , and λ_3 are the tradeoff coefficients. epo denotes the training epoch.

The joint of $\mathcal{L}_R$ and $\mathcal{L}_P$ can exploit the advantages of both reconstruction-oriented and prediction-oriented ideas to realize anomaly detection. However, there exist two problems if only using $\mathcal{L}_R$ and $\mathcal{L}_P$ to optimize FE-CTSNet.

Problem 1: Since only Gaussian functions are adopted to construct fuzzy states, all fuzzy states may become similar during training (the parameters of all Gaussian functions may converge to similar values), which is ineffective to characterize diversiform temporal states.

Problem 2: Since the membership degrees of fuzzy states are randomly initialized before training, fuzzy states at adjacent time points may not have a temporal dependence, which is unable to characterize continuously evolving temporal states.

Therefore, we add two auxiliary objectives $\mathcal{L}_S$ and $\mathcal{L}_T$ to solve problems 1 and 2, respectively. Specifically, $\mathcal{L}_S$ can ensure the differences among fuzzy states to improve its ability to characterize diversiform temporal states. $\mathcal{L}_T$ can endow fuzzy states with temporal dependence to enhance their ability to represent continuously evolving temporal states. Through synthetically co-optimizing $\mathcal{L}_S$, $\mathcal{L}_T$, $\mathcal{L}_R$, and $\mathcal{L}_P$, FE-CTSNet with superior performance can be obtained.

TABLE II
DATASET STATISTICS

	SMD	ASD	MSDS
Train /	68291/	23328/	26357/
Validation set	7589	2592	2929
Test set	75882	12960	29286
Variables	38	19	10

H. Anomaly Inference

In this section, only modules C and D are used to obtain anomaly scores because module B is an auxiliary module to improve the characterization ability of fuzzy states rather than to detect anomalies. The detailed process is summarized in Algorithm 1. It can be seen that each sample sequence $X_{t-T+1:t}^{\text{test}} = \{x_{t-T+1}^{\text{test}}, x_{t-T+2}^{\text{test}}, \dots, x_t^{\text{test}}\} \in \mathbb{R}^{N \times T}$ is input into the well-trained modules C and D to obtain the anomaly score φ for $x_t^{\text{test}} \in \mathbb{R}^N$

$$\phi_{\text{score}}(x_t^{\text{test}}) = \omega_1 \|x_t^{\text{test}} - \tilde{x}_t^{\text{test}}\|_2^2 + \omega_2 \|x_t^{\text{test}} - \hat{x}_t^{\text{test}}\|_2^2 \quad (27)$$

where $\tilde{x}_t^{\text{test}}$ and $\hat{x}_t^{\text{test}}$ denote the reconstructed and predicted values, respectively. Hyperparameters ω_1 and ω_2 are used to tradeoff the reconstruction-oriented error and prediction-oriented error.

Then, peak over threshold (POT) [45] is introduced to adaptively choose the threshold η . If $\varphi(x_t^{\text{test}}) > \eta$, the anomaly label $\hat{y}_t$ is assigned 1 (abnormal), otherwise 0 (normal).

Algorithm 1 FE-CTSNet Anomaly Inference Process

Input: Module C parameter P_C , module D parameter P_D , abnormal and normal test set contains R sample sequences

$X_{1:T}^{\text{test}}, X_{2:T+1}^{\text{test}}, \dots, X_{R:T+R-1}^{\text{test}} \in \mathbb{R}^{N \times T}$ where $X_{1+i:T+i}^{\text{test}} = x_{1+i}^{\text{test}}, x_{2+i}^{\text{test}}, \dots, x_{T+i}^{\text{test}} \in \mathbb{R}^N$ ($0 \leq i \leq R-1$).

Output: All anomaly labels $\hat{y}_{T+i}$ for x_{T+i}^{test} ($0 \leq i \leq R-1$)

- 1: Calculate the threshold η through POT
- 2: **for** $i = 0$ to $R-1$ **do**
- 3: $\tilde{x}_{T+i}^{\text{test}} \leftarrow P_C(X_{1+i:T+i}^{\text{test}})$
- 4: $\hat{x}_{T+i}^{\text{test}} \leftarrow P_D(X_{1+i:T+i-1}^{\text{test}})$
- 5: $\varphi(x_{T+i}^{\text{test}}) \leftarrow \omega_1 \|x_{T+i}^{\text{test}} - \tilde{x}_{T+i}^{\text{test}}\|_2^2 + \omega_2 \|x_{T+i}^{\text{test}} - \hat{x}_{T+i}^{\text{test}}\|_2^2$
- 6: **if** $\varphi(x_{T+i}^{\text{test}}) > \eta$ **then**
- 7: $\hat{y}_{T+i} \leftarrow 1$
- 8: **else**
- 9: $\hat{y}_{T+i} \leftarrow 0$
- 10: **end if**
- 11: **end for**
- 12: **return** all anomaly labels $\hat{y}_{T+i}$ ($0 \leq i \leq R-1$)

IV. EXPERIMENTS

A. Datasets and Evaluation Metrics

To verify the superior performance of FE-CTSNet, we conducted experiments on three public datasets, including server machine dataset (SMD) [24], application server dataset (ASD) [8], and multisource distributed system (MSDS) [12]. In particular, for SMD, we selected machines 1-1, 2-2, and 3-2 as verification objects and reported the average results of these three machines. For ASD, we selected servers 7, 8, and 9 as verification objects and also reported the average results of these three servers. More statistics are shown in Table II.

Moreover, we used four metrics including precision (P), recall (R), F1 score (F1), and area under the receiver operating characteristic curve (AUC) to evaluate the effectiveness of all models. In particular, all the experimental results in this study were average results recorded after five executions.

Currently, many prior studies [1], [24], [46], have calculated the metrics based on a common strategy, i.e., point-adjust: if at least one observation in an anomaly segment is correctly detected as an anomaly, this whole anomaly segment will be regarded as correctly detected. Following this paradigm, the point-adjust strategy was also applied in this study.

B. Baseline Models

1) *State-of-the-Art Models*: To demonstrate our superiority, we selected 14 anomaly detection models as comparison, including IF [4], PCA [18], SFA [20], CVA [21], DAGMM [27], GDN [26], SG-AE [28], OmniAnomaly [24], USAD [29], AAU-Net [25], TranAD [12], TopoMAD [32], MSCRED [30], and MTAD-GAT [31].

2) *Variants of FE-CTSNet*: To show the necessity of each component in FE-CTSNet, we exclude every important one to construct multiple variants, as shown in Table III (each tick denotes the elimination of a component).

C. Model Implementation

FE-CTSNet and all baseline models were implemented using PyTorch on a computer with NVIDIA GeForce RTX 2080 GPU. To ensure the comparison fairness, we adopted POT to automatically select optimal thresholds for FE-CTSNet and all baseline models. Moreover, we also implemented the point-adjust strategy for FE-CTSNet and all baseline models.

For FE-CTSNet, validation sets were used to select the optimal combination of hyperparameters. The final hyperparameters in FE-CTSNet were depicted by the grid search method: the length of sliding window T was 10, the size of ϑ_{ct} was $1 \times 3 \times 1$, and the fuzzy state dimension Q (the number of membership functions) was 150. The number of layers in CTSNet was 3, the number of layers in each mix-hop GCN was 4, the channel dimension C_1 was 64, the channel dimension C_2 was 64, and the number of neurons in the first fully connected layer C_3 was 128. λ_1 , λ_2 , and λ_3 were 0.01, 0.6, and 0.9, respectively. ω_1 and ω_2 were 0.2 and 0.8, respectively. In addition, the training epoch was set to 30 and the batch size was set to 256. In addition, the Adam optimizer with a learning rate of 0.001 is used to conduct end-to-end training.

D. Result Discussion

1) *Overall Performance Comparison*: Table IV reports the anomaly detection results of all models on three datasets. Among machine learning models, IF presents the inferior performance because it neglects temporal and spatial dependencies. In addition, PCA, SFA, and CVA achieve better results due to their consideration of temporal dependence or spatial dependence. However, these models suffer from high false detection rates because their linear hypothesis may not handle complex nonlinear scenarios well.

TABLE III
VARIANTS OF FE-CTSNET

	w/o CSD	w/o TD	w/o FE	w/o $\mathcal{L}_r$	w/o $\mathcal{L}_p$	w/o $\mathcal{L}_t$
Variant 1	✓	-	-	-	-	-
Variant 2	✓	✓	-	-	-	-
Variant 3	-	-	✓	-	-	-
Variant 4	-	-	-	✓	-	-
Variant 5	-	-	-	-	✓	-
Variant 6	-	-	-	-	-	✓
FE-CTSNet	-	-	-	-	-	-

In contrast, deep learning models perform better than machine learning models. However, the detection results of deep learning models show differences. Specifically, DAGMM, SG-AE, GDN, OmniAnomaly, AAU-Net, and TranAD obtain unsatisfactory detection results because they only consider intervariable spatial dependence or intravariation temporal dependence. Compared to the above models, USAD, TopoMAD, MSCRED, and MTAD-GAT gain some promising detection results, demonstrating that the importance of simultaneously considering intervariable spatial dependence and intravariation temporal dependence. Among them, MSCRED is not as reliable as MTAD-GAT and TopoMAD in most metrics because the signature matrix used in MSCRED may fail to identify subtle anomalies and model complex and nonlinear spatial dependence well. Though MTAD-GAT and TopoMAD use the graph structure to realize the relatively great performance compared with other baseline models, it still remains suboptimal performance because it ignores cross-time spatial dependence. In summary, FE-CTSNet achieves the best F1 and AUC on SMD and ASD, and the best F1 and the second-best AUC on MSDS, which illustrates the importance of disentangling mixing temporal states and capturing cross-time spatial dependence.

To further support the conclusion above, Fig. 8 visualizes the anomaly detection results of FE-CTSNet and some deep learning models. Gray solid lines and red dashed lines represent anomaly scores and threshold, respectively. Blue-shaded regions represent the predicted anomalies, and red-shaded regions represent the true anomalies, which contain two types: pointwise anomalies and collectivewise anomalies. In all models, the observations whose anomaly scores exceed the corresponding threshold are marked as anomalies. For SMD (machine 1-1), all models can identify all anomalies. However, some deep learning models present many false alarms (e.g., GDN, USAD, MTAD-GAT, and OminAnomaly), which is not conducive for practical scenarios. For ASD (server 7), compared to other baseline models, FE-CTSNet can generate higher anomaly scores for true anomalies and smoother anomaly scores for normal moments, which demonstrates that FE-CTSNet is more sensitive and robust to the evolution laws of time series.

2) *Ablation Study*: This section investigates the effectiveness of each important component in FE-CTSNet. The relevant variants are listed in Table II, and the corresponding detection results from dataset ASD (server 7) are shown in Fig. 9. We can draw the following conclusions.

TABLE IV
PERFORMANCE COMPARISON OF FE-CTSNET WITH THE STATE-OF-THE-ART MODELS ON THREE DATASETS

Model	SMD				ASD				MSDS			
	P	R	F1	AUC	P	R	F1	AUC	P	R	F1	AUC
IF	0.1662	0.9985	0.2810	0.7618	0.1044	0.9280	0.1705	0.7189	0.7644	0.9888	0.8623	0.5972
PCA	0.2340	1.0000	0.3612	0.8021	0.2061	1.0000	0.3322	0.9049	0.8379	0.9985	0.9112	0.7477
SFA	0.4179	0.9985	0.5891	0.9323	0.1950	1.0000	0.3171	0.9007	0.9759	0.9308	0.9528	0.9355
CVA	0.4460	0.9985	0.5743	0.8978	0.4578	0.9031	0.5953	0.9143	0.9997	0.7986	0.8879	0.8990
DAGMM	0.6564	0.8359	0.6857	0.8864	0.6928	0.8185	0.7296	0.8991	0.9973	0.5486	0.6491	0.7708
SG-AE	0.9004	0.9578	0.9204	0.9717	0.8876	0.8205	0.8461	0.9059	0.9995	0.8207	0.9009	0.9097
GDN	0.8848	0.9317	0.8957	0.9571	0.9327	0.7339	0.8054	0.8654	1.0000	0.7218	0.8351	0.8609
OmniAnomaly	0.8895	0.9364	0.9038	0.9605	0.9325	0.7985	0.8564	0.8977	1.0000	0.7964	0.8867	0.8982
AAU-Net	0.6973	0.9738	0.7884	0.9663	0.8071	0.8933	0.8449	0.9424	0.9999	0.8457	0.9163	0.9228
TranAD	0.9588	0.9172	0.9157	0.9561	0.9363	0.7776	0.8347	0.8877	0.9976	0.8138	0.8964	0.9044
USAD	0.8980	0.9553	0.9213	0.9713	0.9311	0.3896	0.4557	0.6945	0.9963	0.8818	0.9345	0.9363
TopoMAD	0.9305	0.9216	0.9217	0.9563	0.8093	0.8754	0.8379	0.9313	0.9986	0.9135	0.9541	0.9551
MSCRED	0.8593	0.9030	0.8652	0.9413	0.5235	0.9034	0.6505	0.9294	0.9956	0.8840	0.9356	0.9367
MTAD-GAT	0.9434	0.9603	0.9489	0.9766	0.8134	0.9155	0.8587	0.9522	0.9999	0.8026	0.8904	0.9012
FE-CTSNet	0.9550	0.9644	0.9575	0.9794	0.9275	0.9556	0.9398	0.9756	0.9893	0.9242	0.9548	0.9483

* SIGNIFICANT VALUES ARE BOLDFACED.

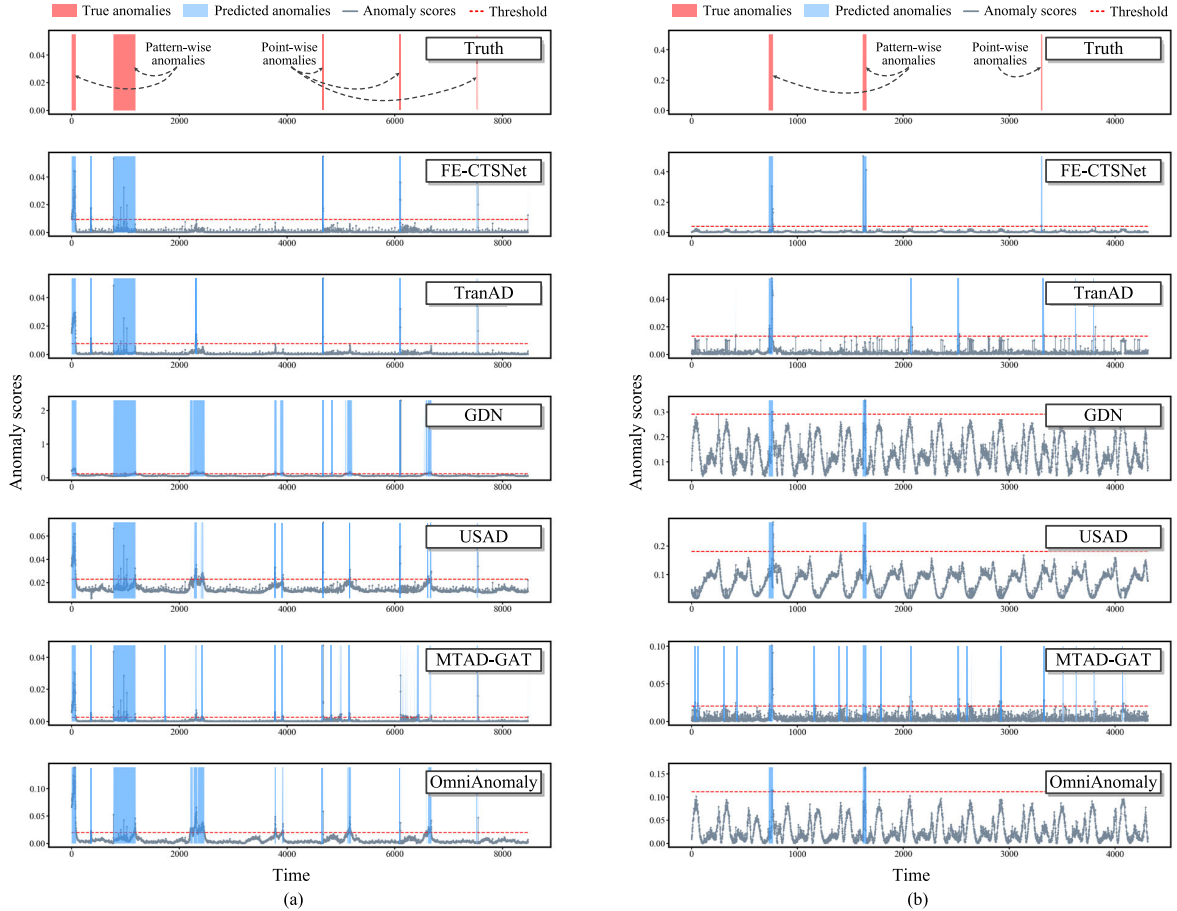


Fig. 8. Predicted and true anomalies on SMD (machine 1-1) and ASD (server 7). Since collectivewise anomalies is a contiguous anomaly segment, the point-adjust method is adopted: if at least one observation in collectivewise anomalies is predicted as an anomaly, the whole collectivewise anomalies will be regarded as correctly detected. Therefore, some observations in collectivewise anomalies with anomaly scores below the threshold are also labeled to be anomalies. (a) SMD (machine 1-1). (b) ASD (server 7).

- 1) *Compare FE-CTSNet With Variants 1 and 2*: The ignorance of cross-time spatial dependence and temporal dependence can both remarkably degrade the model performance.
- 2) *Compare FE-CTSNet With Variant 3*: Module A (fuzzy embedding module) is superior to fully connected layers, which indicates that using fuzzy states to characterize latent and mixing temporal

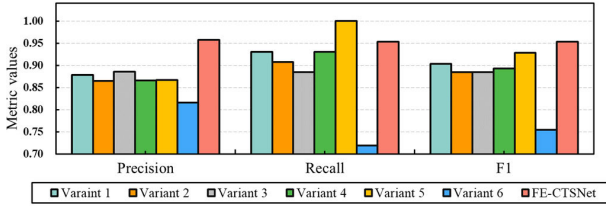


Fig. 9. Ablation study results on ASD (server 7).

states is helpful for improving anomaly detection results.

- 3) *Compare FE-CTSNet With Variants 4, 5, and 6*: Modules B, C, and D are all essential for FE-CTSNet. Moreover, module D has a more significant contribution to model performance than the other two modules, illustrating the potential of CTSG in capturing complete spatiotemporal dependencies for promoting detection results.

E. Specific Analysis

1) *Effectiveness of CTSG*: Fig. 10 visualizes an example of CTSG after training. The horizontal coordinate and vertical coordinate in the heatmap indicate the corresponding variable indexes (ten variables at time point t_1 and ten variables at time point t_2). A cell with a darker color indicates the stronger interaction between pairwise variables, while a cell with a lighter color indicates the opposite. As can be seen, CTSG can uniformly and flexibly characterize complete spatiotemporal dependencies. Specifically, CTSG captures strong autocorrelations of variables along the time axis (time interval 1), as shown in the green area, which proves the effectiveness of the designed temporal proximity constraint in ensuring the temporal dependence of fuzzy states. Moreover, it can also capture other intervariable correlations, i.e., intervariable correlations at the same time point in the gray area and intervariable correlations at different time points in the orange area. These intervariable correlations are crucial for discovering anomalies in multivariable time series because anomalies often propagate between different variables.

2) *Effectiveness of Fuzzy States*: Fig. 11 provides a real server cluster example (a server is denoted as a variable) of how FE-CTSNet exploiting fuzzy states to detect the time-delay anomaly. Specifically, the time-delay anomaly propagates from abnormal time period A in variable 2 to abnormal time period B in variable 8. Through FE-CTSNet, each data point in variables 2 and 8 is converted into a corresponding membership vector. Apparently, membership vectors exhibit difference between abnormal time points and normal time points (e.g., membership vectors at normal time point 755 and abnormal time point 758 in variable 2), indicating that fuzzy states have the ability to distinguish between anomaly and normal. Further, we find out that the membership vectors of variable 2 at anomaly time period A are quite similar to that of variable 8 at anomaly time period B, indicating that fuzzy states successfully discovers strong cross-time spatial dependencies between them, thereby endowing FE-CTSNet with the ability to sense the time-delay anomaly propagating from variable 2 to variable 8. Based on it, if the anomaly

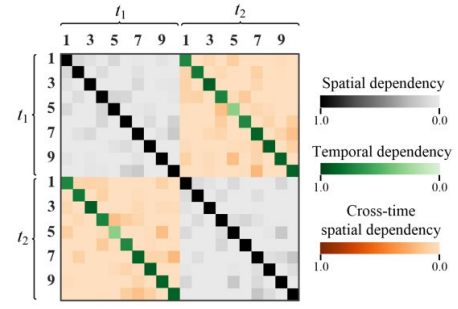


Fig. 10. Example of CTSG in the validation set on MSDS.

in abnormal time period A can be detected, FE-CTSNet will easily infer the time-delay anomaly in abnormal time period B.

3) *Hyperparameter Test*: We evaluate the sensitivities of some key hyperparameters in FE-CTSNet: tradeoff coefficients in the joint evolution objective $\mathcal{L}_J$, the number of layers in CTSNet S , and the number of fuzzy states Q .

a) *Sensitivity of λ_1 and λ_2 in $\mathcal{L}_J$* : λ_1 and λ_2 control the influences of two auxiliary objectives (i.e., state diversity constraint and temporal proximity constraint) on $\mathcal{L}_J$. Too small λ_1 and λ_2 may make fuzzy states lose their diversity and evolution continuity, thus degrading the ability to characterize latent temporal states. Too large λ_1 and λ_2 may hamper the normal optimization of main objectives (prediction and reconstruction for time series), leading to poor anomaly detection performance. We test λ_1 with values from $\{0.001, 0.01, 0.1, 0.5, 1\}$ and λ_2 with values from $\{0.2, 0.4, 0.6, 0.8, 1\}$. It can be found that the superior results are gained when $\lambda_1 = 0.01$ and $\lambda_2 = 0.6$ [Fig. 12(a) and (b)], which is consistent with the previous theoretical analysis.

b) *Sensitivity of λ_3 in $\mathcal{L}_J$* : λ_3 controls the balance between the prediction task and the reconstruction task. Fig. 12(c) shows that λ_3 with large value (0.9) realizes the best performance, indicating that the joint evolution objective should focus more on the reconstruction module in the early training phase.

c) *Sensitivity of S* : S controls the ability of FE-CTSNet to capture long-term cross-time spatial dependence. As shown in Fig. 13(a) (solid lines and shadows indicate the average results and the standard deviations, respectively), it is found that $S = 3$ performs best in terms of all metrics, illustrating that too small S may fail to effectively capture long-term spatiotemporal dependencies, while too large S may lead to overfitting problems.

d) *Sensitivity of Q* : Q controls the number of fuzzy states for disentangling mixing temporal states. As shown in Fig. 13(b), it is found that using $Q = 150$ obtains better detection results than those obtained by using other values in terms of the maximum, minimum, and median of F1. This illustrates that a certain number of fuzzy states is necessary to reasonably characterize latent and mixing temporal states, but redundant fuzzy states may degrade its characterization ability.

4) *Convergence of $\mathcal{L}_J$* : To demonstrate the reasonability of the optimization process in FE-CTSNet, the losses of $\mathcal{L}_J$, $\mathcal{L}_S$, $\mathcal{L}_T$, $\mathcal{L}_R$, and $\mathcal{L}_P$ in the validation set are shown in Fig. 12(d). The loss of $\mathcal{L}_J$ quickly reaches convergence with small fluctuations, which verifies that all modules of

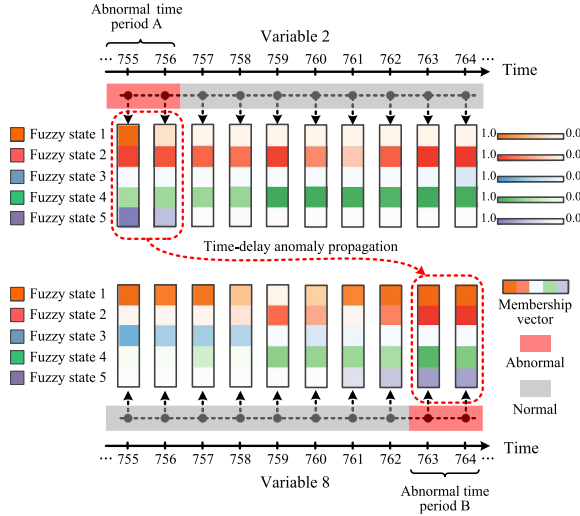


Fig. 11. Five fuzzy states in the test set on ASD (server 9) are randomly chosen as an example to illustrate its effectiveness. The color of each cell in the membership vector represents the magnitude of the membership degree (a cell with a darker color indicates the larger membership degree).

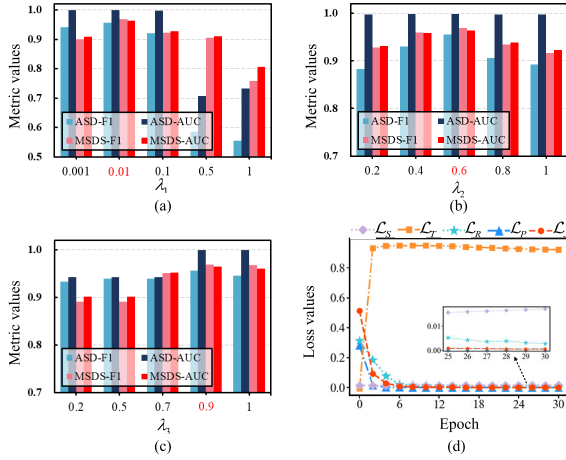


Fig. 12. Hyperparameter test results. (a)–(c) F1 and AUC of λ_1 , λ_2 , and λ_3 on ASD (server 7) and MSDS. (d) Convergence analysis in the validation set on ASD (server 9).

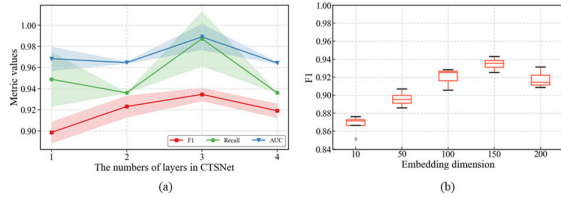


Fig. 13. Hyperparameter test results. (a) and (b) Sensitivity analysis of S and Q on ASD (server 9).

FE-CTSNet can be co-optimized well during the training iterations. Simultaneously, the losses of $\mathcal{L}_S$ and $\mathcal{L}_T$ rise gradually at the beginning of training and then become stable, which verifies that the $\mathcal{L}_S$ and $\mathcal{L}_T$ can effectively realize their respective goals, that is, to enhance the diversity and evolution continuity of fuzzy states; the loss of $\mathcal{L}_R$ and $\mathcal{L}_P$ drops gradually during the training iterations, illustrating that FE-CTSNet can successfully capture complete spatiotemporal dependencies (especially, cross-time spatial dependence) and learn specific patterns of time series to reconstruct and predict time series, thereby achieving effective anomaly detection.

V. CONCLUSION

In this study, we propose a novel FE-CTSNet to realize the disentanglement for mixing temporal states from a fuzzy embedding perspective and exploiting it to sense the intricateness and mutability of cross-time spatial dependence, thus improving the detection performance for time-delay anomalies. Specifically, the proposed TS2Mvec can build fine-grained fuzzy states to represent latent temporal states and generate membership degrees to accurately reflect mixing modes and dynamical evolutions of temporal states. Built upon fuzzy states, the proposed CTSG can flexibly learn mutable cross-time spatial dependence. Moreover, the designed state diversity and temporal proximity constraints possess the great convergence, which can effectively ensure the differences among fuzzy states and the evolution continuity of fuzzy states, thereby further enhancing their characterization ability for latent temporal states. Experiments on three public datasets demonstrate the superiority of FE-CTSNet and the effectiveness of its important components. In addition, the visualization of fuzzy states and CTSG illustrates the excellent potential of FE-CTSNet in discovering time-delay anomalies.

Future directions will focus on distinguishing transient impulsive interferences and anomalies, enabling FE-CTSNet to achieve more fine-grained and accurate anomaly detection in practical industrial applications.

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